# SuperScribble: A Super-Pixel and Pixel Dual Constraint Remote Sensing Image Semantic Segmentation Network With Scribble Annotation

Anqi Zhao, Hongjin Wang, Xinghua Li, *Senior Member, IEEE*, and Huanfeng Shen, *Senior Member, IEEE*

***Abstract*—Semantic segmentation is a crucial task in remote sensing field. With the rapid advancement, most learning-based methods rely heavily on pixel-level annotation data for training. To reduce annotation burden, more and more weakly supervised semantic segmentation (WSSS) methods are developed. Among these WSSS annotations, scribble is flexible and easily obtainable during practical labeling. Since there are relatively few multiclass WSSS methods for remote sensing images (RSIs) based on scribble annotation, a super-pixel and pixel dual constraint RSI semantic segmentation network with scribble annotation (SuperScribble) is proposed. SuperScribble consists of three models: super-pixel feature extraction (SFE) model, pixel feature extraction (PFE) model, and multihead attention feature fusion (MAF) model. SFE and PFE are designed to extract super-pixel features and pixel features, respectively. On the one hand, they jointly generate more accurate pseudo labels. On the other hand, they are fused in MAF to achieve better segmentation. To further guarantee the quality of pseudo labels, a dual constraint strategy (DCS) for pseudo label generation is adopted, which is proved that it's highly compatible with dual-branch networks. Experimental results demonstrate that compared with other eight methods, SuperScribble could achieve competitive results on three public datasets, namely Potsdam, Vaihingen, and Zurich Summer datasets. The implementation code and data will be available at: https://github.com/lixinghua5540/SuperScribble**

***Index Terms*—Remote sensing images (RSIs), scribble annotation, semantic segmentation, weakly supervised.**

## I. Introduction

SEMANTIC segmentation aims to label each pixel on remote sensing image (RSI) with the correct class label, which is playing an increasingly significant role in key areas such as resource monitoring [1], [2], land use/cover mapping [3], [4], damage assessment [5], [6], and so on. More and more researchers have attempted various deep learning networks to achieve semantic segmentation for RSI. The existing deep learning networks mainly rely on large amounts of accurate pixel-level annotations for fully supervised semantic segmentation (FSSS) training [7], [8], [9], [10], [11]. However, it requires assigning right category to all pixels in RSI, which is time-consuming and labor-intensive.

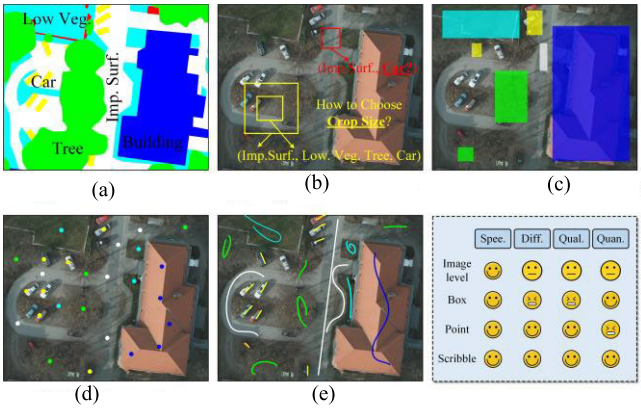



Fig. 1. Annotation from different semantic segmentation methods. "Spee." denotes annotation speed, "Diff." denotes annotation difficulty, "Qual." denotes annotation quality, and "Quan." denotes annotation quantity. (a) FSSS annotation. (b) Image-level annotation. (c) Bounding box annotation. (d) Point annotattion. (e) Scibble annotation.

To reduce annotation burden, researchers have developed weakly supervised semantic segmentation (WSSS) methods for RSI, aiming to gain satisfactory results with less annotation data. WSSS methods adopt incomplete or imprecise labels for training, such as image-level [12], [13], bounding box [14], point [15], [16], [17], scribble [18], [19], and so on. As Fig. 1 presented, although these methods have reduced annotation burden, there are still some problems. For image-level annotation, it's a dilemma to choose size of each annotated image, which affects the difficulty and quality of the labels. For example, when annotating the red boxed regions in Fig. 1, it is difficult to determine whether cars are present in the imagery. When the annotated regions are excessively large, such as the yellow boxed areas in Fig. 1, each image tends to contain all semantic classes, thus reducing the discriminative value of image-level labels. For bounding box annotation, it's imprecise due to irregular shape of objects in RSI. Point annotation is small, which provides insufficient information for the subsequent network training. After comparison, the application of scribble annotation in WSSS is promising. It's quick and easy to obtain sufficient scribble annotation. In remote sensing domain, scribble annotation has been widely used for extraction of a certain type of land objects, such as

Received 24 January 2026; revised 6 April 2026; accepted 28 April 2026. Date of publication 4 May 2026; date of current version 11 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42571368. *(Corresponding author: Xinghua Li.)*

Anqi Zhao, Hongjin Wang, and Xinghua Li are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: zhaoanqi@whu.edu.cn; wanghj@whu.edu.cn; lixinghua5540@whu.edu.cn).

Huanfeng Shen is with the School of Resource and Environmental Science, Wuhan University, Wuhan 430079, China (e-mail: shenhf@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3689583

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

roads or buildings [20], [21], [22]. However, for multiclass segmentation tasks, there are relatively few WSSS methods based on scribble annotation. Therefore, this study focuses on multiclass WSSS method based on scribble annotation for RSI.

Existing WSSS methods mainly focus on two aspects to improve the segmentation accuracy: generating accurate pseudo labels and improving network performance. Recently, WSSS methods take attentions on outputs from different branches to generate more reliable pseudo labels. For example, RLSP [13] introduces two reinforcement learning agents to generate more accurate pseudo labels by competing between agents. ScribFormer [18] employs dual branches to generate pseudo labels by average of two outputs from convolutional neural networks (CNNs) and Transformer. SparseFormer [17] adopts two different CNN to jointly generate pseudo labels with high confidence. These methods all expect to produce high-quality pseudo labels by comprehensively leveraging features from different branches. The diversity of features from different branches helps to ensure the reliability of pseudo labels.

Super-pixel and pixel are vision tokens with different styles. Super-pixel could perform geometric modeling of irregular object structures [23], [24], and pixel-level representations preserve fine-grained details. The combination of super-pixel and pixel could enrich feature diversity, which is a promising approach for WSSS in RSI. Therefore, a super-pixel and pixel dual constraint RSI semantic segmentation network with scribble annotation (SuperScribble) is proposed. SuperScribble consists of three models: super-pixel feature extraction (SFE) model, pixel feature extraction (PFE) model and multihead attention feature fusion (MAF) model. The main contributions of this work are emphasized as follows.

1) Scribble annotations of three public high-resolution datasets are constructed, namely Potsdam, Vaihingen, and Zurich Summer dataset. They offer valuable benchmarks for evaluating different WSSS methods based on scribble annotation in RSI.
2) SuperScribble is proposed, a novel method for WSSS of RSI with scribble annotation. SuperScribble introduces SFE and PFE to rich feature representation, which extract global and local features with different styles, respectively. These features are further fused in MAF, achieving better segmentation results from coarse to fine.
3) A dual constraint strategy (DCS) for pseudo label generation and a staged training strategy are adopted. DCS strictly controls the generation of pseudo labels, which avoid the interference of false labels for training. Staged training strategy not only enhances the quality of pseudo labels but also improves the segmentation accuracy.

## II. Related Work

Semantic segmentation is an important task in the field of remote sensing. With the prosperity of deep learning, a large number of learning-based methods have been developed. The scribble annotation-based method achieves learning WSSS, which aims to reduce annotation burden from FSSS methods. Consequently, this section will provide a detailed description of the current advancements in FSSS and WSSS. In addition, DCS is also one of the main contributions in this article; therefore, it will also introduce the current pseudo label generation strategy.

### A. Fully Supervised Semantic Segmentation

Many researchers design various network structures, such as CNN [9], [25], [26], fully convolutional network (FCN) [7], generative adversarial networks (GANs) [27], [28], and U-Net [8], [29]. These methods adopt learnable network frameworks and incorporate various novel modules to enhance the segmentation performance. For example, classical SegNet [25] adopts a structure of encoder and decoder to achieve excellent performance with light parameters. BEDSN [9] introduces an edge detection branch to compensate for boundary loss. MANet [26] employs multiple efficient attention modules to fuse low-level features and high-level features.

Recently, it has realized that transformer is better to extract global features than CNN; therefore, a lot of methods based on transformer are presented [8], [30], [31]. For example, UNetFormer [8] develops an efficient global–local attention mechanism to model both global and local information. TMFNet [31] introduces digital surface model to supplement the extra features, which are enhanced by transformer.

Although these methods based on deep learning gain amazing segmentation results, they rely on large amounts of accurate pixel-level annotations for FSSS. Obtaining enough training labels for FSSS is quite time-consuming, which also poses great challenges for annotators in practice.

### B. Weakly Supervised Semantic Segmentation

To reduce annotation burden, WSSS methods for RSI are proposed, these methods adopt incomplete or imprecise labels for training, such as image-level [12], [13], [32], bounding box [14], point [15], [16], [17], scribble [18], [19], and so on. Image-level annotation only needs to point out whether a certain type of geographical objects exists in the image, without location, extent, or shape information. For example, OME [12] is trained to form class activation maps, representing probabilities maps, which further generate pseudo labels for subsequent training. But as emphasized, size of each image-level annotation requires manual parameter adjustment, which further affects the cost and quality of the annotations. In addition, some methods conduct WSSS by oriented bounding box such as OBBInst [14]. Bounding box annotation is quick, but shapes of the objects in RSI are diverse, which leads to annotation quality being poor.

To further reduce annotation burden and guarantee annotation quality, WSSS in RSI supervised by point annotation are gradually favored, such as CRGNet [15] and SparseFormer [17]. CRGNet introduces region-growing mechanism and dual branch VGG16 to guarantee the quality of pseudo labels. Point annotation is not only quick but also precise, but due to sparse and small, it provides insufficient information for network training. In terms of scribble annotation, it's quick and easy to obtain precise and sufficient scribble due to its flexibility. In natural images, a number of scribble-based WSSS methods have been proposed, such as ScribFormer [18] and CC4S [19].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

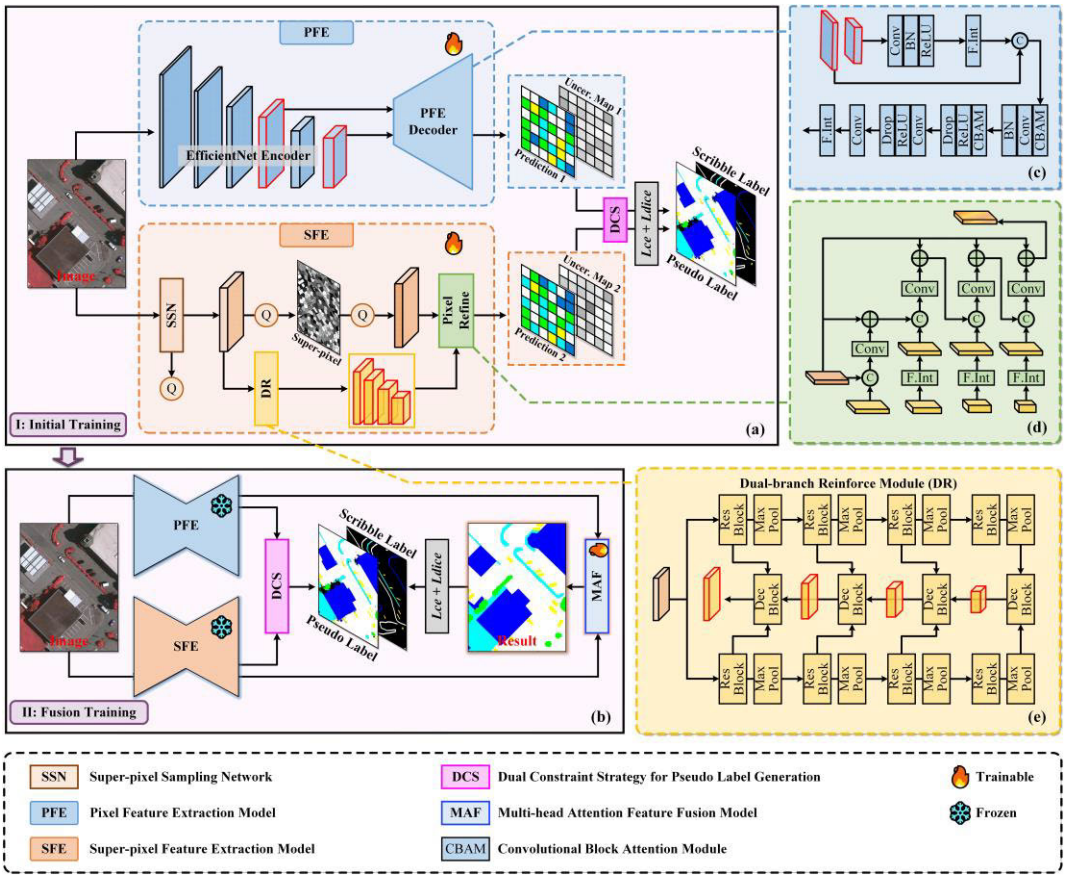



Fig. 2. Overall architecture of proposed SuperScribble.

CCS4 incorporates dual branches with different supervision to enhance feature. In RSI, scribble annotation has been used for extraction of a certain type of land objects, such as road extraction [20], building extraction [21] and object detection [22]. However, for multiclass semantic segmentation tasks in RSI, WSSS methods based on scribble annotation are barely explored.

### C. Pseudo Label Generation

As the core of WSSS, quantity and quality of pseudo label both greatly affect the subsequent segmentation accuracy. Some methods take the model's predictions with confidence maps as pseudo labels [33], [34]. For example, according to outputs from VGG16, CRGNet [15] uses the one with the largest probability as the pseudo label. This strategy doesn't impose any constraints on the uncertain results, which generates a lot of false labels. Some methods refine the predictions to ensure the reliability of pseudo labels. For instance, the uncertainty maps are introduced to improve pseudo labels [35]. Some methods emphasize the importance of iterative refinement and cross-view consistency for pseudo label generation, such as IRNet [36]. IRNet leverages interpixel relations to propagate reliable labels. In addition, AGMM [37] assumes that there are sparse annotated point labels to represent the existing every object, and it improves predictions based on such assumption. These approaches further highlight that high-quality pseudo labels are essential for stable and accurate training.

Meanwhile, some researchers adopt different branches to output features with different styles, which aim to gain more accurate pseudo labels by feature variety, such as ScribFormer [18], RLSP [13] and BACF [38]. In addition, SparseFormer [17] takes two different CNN to jointly generate pseudo labels by competitive mechanism, namely prediction results with high probability are regarded as pseudo labels. Although these methods enhance constrains to pseudo labels, the accuracy of pseudo labels needs to be improved. In terms of pseudo label generation, it's better not to generate uncertain labels to mislead subsequent network training.

## III. Methodology

### A. Overall Architecture of SuperScribble

The overall architecture of SuperScribble is displayed in Fig. 2. As Fig. 2 shown, the training process of SuperScribble is divided into two stages: initial training and fusion training. At the stage of initial training, for diversity of features, PFE and SFE are separately trained at the supervision of origin scribble label. After that, pre-trained PFE and SFE could generate pseudo labels through DCS, and PFE and SFE are further trained at the supervision of scribble labels and pseudo labels. At the stage of fusion training, frozen PFE and SFE also first generate pseudo labels through DCS, and MAF are then trained at the supervision of scribble label and pseudo labels, which further fuse features from frozen PFE and SFE. SuperScribble mainly consists of PFE, SFE, DCS, and MAF, and they will be elaborately explained in the following.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

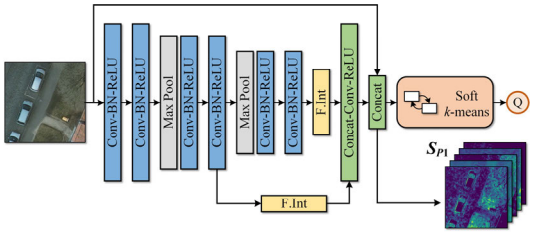



Fig. 3. Architecture of SSN.

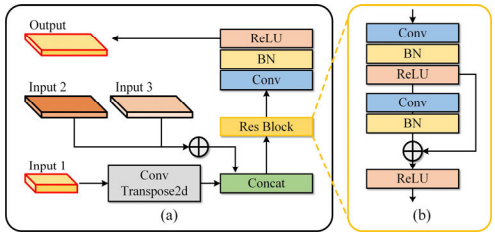



Fig. 4. Architecture of some modules in DR. (a) Dec block. (b) Res block.

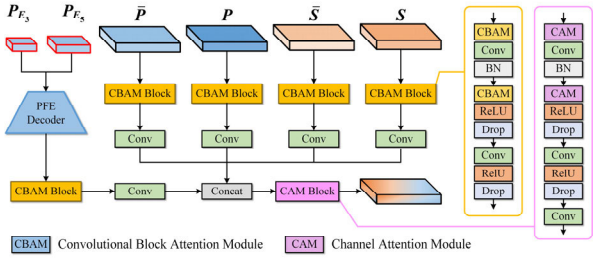



Fig. 5. Architecture of MAF.

## B. PFE Model

Aiming to extract pixel features, namely local information, the encoder of PFE is designed based on EfficientNet [39], which gains feature maps with different resolutions as follows:

$$P_{E_i} = EE(I) \tag{1}$$

where $I$ represents image with the dimension of $H \times W \times 3$, $EE$ denotes EfficientNet encoder and $P_{E_i}(i = 0, 1 \ldots, 5)$ denotes encoder features output by $EE$, whose spatial resolution are $((H/2^i), (W/2^i))$, respectively. $P_{E_{i+1}}$ is obtained through $P_{E_i}$ and corresponding EfficientNet block.

As Fig. 2(c) presented, PFE decoder takes $P_{E_3}$ and $P_{E_5}$ as input, which consists of convolution layer, batch normalization layer, ReLU activation function, bilinear interpolation function, and convolutional block attention module (CBAM) [40]. There are two interpolation functions in PFE decoder, and the spatial size of features output by the first and the second are same as $P_{E_3}$ and $P_{E_0}$, respectively. The final feature $P$ output by PFE decoder is as follows:

$$\bar{P} = PD(P_{E_3}, P_{E_5}) \tag{2}$$

$$P = Conv(\bar{P}) \tag{3}$$

where $PD$ denotes PFE decoder, the dimensions of $\bar{P}$ and $P$ are $H \times W \times C_1$ and $H \times W \times K$, respectively, $K$ denotes the number of land use categories.

## C. SFE Model

With the purpose of extracting super-pixel features, namely global information, SFE is designed based on SSN [23], [41]. Fig. 3 shows the architecture of SSN, which obtains initial pixel features $S_{P1}$ and $Q$ as follows:

$$S_{P1}, Q = SSN(I) \tag{4}$$

where the dimension of $S_{P1}$ is $N \times C_1$, $N = H \times W$. $Q$ represents soft association matrix between pixel features and superpixel features, whose calculation is an iterative process. Specifically, $S_{P1}$ is used in the differentiable simple linear iterative clustering [42] for the iterative calculation of $Q$. In addition the dimension of $Q$ is $N \times N_S$, which indicates the probability that each pixel belongs to every super-pixel and $N_S$ denotes the number of super-pixels. Super-pixels themselves mainly capture regional structures. However, since each pixel is associated with all super-pixels through $Q$, this dense interaction enables each pixel to aggregate information from multiple regions across the entire image, allowing the network to capture global information.

By using the column-normalized $Q$, denoted as $\hat{Q}$, and the row-normalized $Q$, denoted as $\tilde{Q}$, super-pixel features and enhanced pixel features could be obtained as follows:

$$S_s = \hat{Q}^T S_{P1} \tag{5}$$

$$S_{P2} = \tilde{Q} S_s \tag{6}$$

where $S_s$ denotes superpixel features with the dimension of $N_S \times C_2$, $S_{P2}$ denotes enhanced pixel features with the dimension of $N \times C_2$. For readers, more details about SSN could refer to related works [23], [41].

Meanwhile, in order to further refine features in SFE, another module with $S_{P1}$ as input is introduced as Fig. 2(e) shown, named dual-branch reinforce (DR) module. DR contains Res block and Dec block, whose architectures are displayed in Fig. 4. The "input 1" in Fig. 4 performs upsampling through transposed convolution. The DR outputs features with four different resolutions as follows:

$$S_{D_i} = DR(S_{P1}) \tag{7}$$

where $S_{D_i}$ $(i = 0, 1, 2, 3)$ denotes features output by DR, whose spatial resolution is $((H/2^i), (W/2^i))$, respectively.

Finally, $S_{P2}$ and $S_{D_i}$ are fused through pixel refine module as Fig. 2(d) shown, which is a process of multiscale features fusion. The final feature $S$ output by SFE is as follows:

$$\bar{S} = PR(S_{P2}, S_{D_i}) \tag{8}$$

$$S = Conv(\bar{S}) \tag{9}$$

where $PR$ represents pixel refine operation and the dimensions of $\bar{S}$ and $S$ are $H \times W \times C_3$ and $H \times W \times K$, respectively.

## D. MAF Model

To further fuse features from PFE and SFE, MAF is proposed. MAF takes some features from frozen pretrained PFE and SFE as input, which includes $P_{E_3}$, $P_{E_5}$, $\bar{P}$, $P$, $\bar{S}$, and $S$. Fig. 5 presents the architecture of MAF, which is based on PFE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

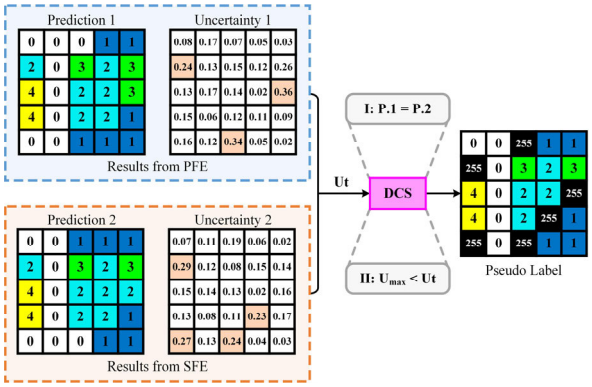



Fig. 6. Sample of DCS. In pseudo label categories, "255" denotes invalid label.

decoder, CBAM block, and channel attention module (CAM) block [40]. For inputs from PFE and SFE, there exits five CBAM blocks in MAF to enhance them. Finally, considering that these features contribute unequally to the segmentation, CAM block is introduced. The final feature output by MAF is as follows:

$$M = MAF(P_{E_3}, P_{E_5}, \bar{P}, P, \bar{S}, S) \tag{10}$$

where $M$ denotes final feature output by MAF with the dimension of $H \times W \times K$.

### E. DCS for Pseudo Label Generation

Aiming to generate high-quality pseudo labels, DCS is adopted. Through a careful and comprehensive evaluation for the outputs from the SFE and PFE, DCS strictly controls the generation of pseudo labels. Specifically, DCS includes two operations. First, the prediction results $P_1$ and $P_2$ from PFE and SFE must be consistent. Second, the uncertainty results $U_1$ and $U_2$ from PFE and SFE are must less than the set threshold $U_t$. The larger $U_t$ is, the more pseudo labels there will be, and the lower the accuracy of the pseudo labels will be. Fig. 6 shows a sample of DCS. The process in DCS could be expressed as follows:

$$\bar{P}_p = \begin{cases} p_1(i,j), & P_1(i,j) = P_2(i,j) \\ 255, & P_1(i,j) \neq P_2(i,j) \end{cases} \tag{11}$$

$$P_p = \begin{cases} \bar{P}_p(i,j), & U_{max}(i,j) < U_t \\ 255, & U_{max}(i,j) \geq U_t \end{cases} \tag{12}$$

$$U_{max} = max(U_1, U_2) \tag{13}$$

where $P_1$ and $P_2$ denotes the category with the highest probability in $P$ and $S$, respectively. $(i, j)$ denotes coordinates of pixels, and 255 denotes the category that do not participate in the segmentation. $P_p$ denotes final pseudo labels generated by DCS.

Inspired by previous work [17], uncertainty results $U_1$ and $U_2$ from PFE and SFE are calculated based on $P$ and $S$ respectively, which are as follows:

$$U_1 = -(P_{max} log(P_{max}) + P_{sec} log(P_{sec})) \tag{14}$$

$$U_2 = -(S_{max} log(S_{max}) + S_{sec} log(S_{sec})) \tag{15}$$

where $P_{max}$ and $P_{sec}$ denote the predicted maximum and second maximum probabilities in $P$, respectively. $S_{max}$ and $S_{sec}$ denote the predicted maximum and second maximum probabilities in $S$, respectively.

### F. Training Strategy and Loss Function

A staged training strategy is introduced for SuperScribble. There are two stages in training of SuperScribble. First, PFE and SFE are separately trained at the supervision of original scribble labels. Then, pretrained PFE and SFE could generate pseudo labels through DCS, and then PFE and SFE are further separately trained at the supervision of scribble labels and pseudo labels. This stage is iterative. Finally, MAF is trained at the scribble labels and pseudo labels, which takes some features from frozen PFE and SFE as inputs. At inference stage, an input image is first fed into the PFE and SFE to obtain features. These features are then fused by MAF, which produces the final segmentation prediction.

Based on previous works [8], [17], cross-entropy loss function and Dice loss function are effective in WSSS, which are used for SuperScribble's training. First, at the supervision of original scribble labels, loss function for PFE and SFE is as follows:

$$L_s = L_{ce}(\hat{y}, y_s) + L_{dice}(\hat{y}, y_s) \tag{16}$$

where $L_s$ denotes the loss for scribble labels, $L_{ce}$ and $L_{dice}$ denote cross-entropy and Dice loss function respectively. $\hat{y}$ and $y_s$ represent prediction results and scribble labels, respectively. Next, after pseudo labels are generated, loss for them is added, which is as follows:

$$L_p = L_{ce}(\hat{y}, y_p) + L_{dice}(\hat{y}, y_p) \tag{17}$$

$$L = \lambda_1 L_s + \lambda_2 L_p \tag{18}$$

where $L_p$ denotes the loss for pseudo labels, and $y_p$ denotes pseudo labels. $\lambda_1$ and $\lambda_2$ denote loss weights. $L$ represents the loss for subsequent PFE, SFE and MAF.

## IV. Experiment Results

### A. Network Settings and Experimental Data

SuperScribble consists of PFE, SFE, and MAF. The training settings for PFE, SFE, and MAF are consistent. Specifically, the network iterates 6000 times with a learning rate of $1 \times 10^{-4}$. Adam optimizer is introduced with betas (0.5, 0.999). The image size is $256 \times 256$ for training. In experiments, $U_t = 0.1$, $\lambda_1 = \lambda_2 = 0.5$. For PFE, EfficientNet-B3 is used as encoder. The SuperScribble is implemented on the PyTorch framework. The experimental platform is NVIDIA Quadro RTX 5000 GPU with 16 GB of memory.

The experiments are conducted based on three public datasets, namely Potsdam, Vaihingen, and Zurich Summer datasets. Although Hua et al. [43] have shared point and scribble annotation for Vaihingen and Zurich Summer datasets, they are too sparse as Fig. 7 shown. Therefore, scribble annotations for three datasets are constructed by authors with the help of labelme software as Fig. 8 displayed. Table I shows the number of annotations for three datasets, which include original full pixelwise annotations and constructed scribble

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

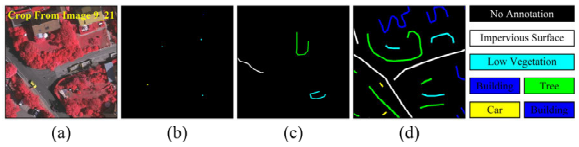



Fig. 7. Annotations for Vaihingen dataset. (a) Image. (b) Point annotation, existing. (c) Scribble annotation, existing. (d) Scribble annotation, constructed.

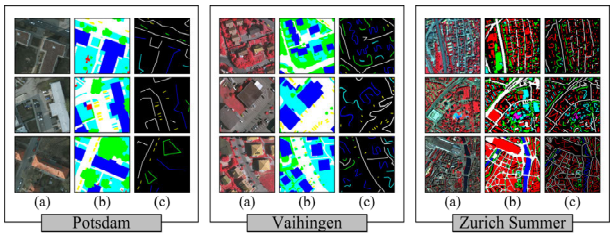



Fig. 8. Samples of constructed scribble annotations. (a) Image. (b) FSSS annotation. (c) Constructed scribble annotation.

TABLE I
BRIEF INFORMATION OF THREE DATASETS

| Dataset | Potsdam | Vaihingen | Zurich Summer |
|---|---|---|---|
| Resolution | 0.05 m | 0.09 m | 0.62 m |
| Size | 6000×6000 | 2500×1900 | 1000×1000 |
| Full annotation | 791,663,578 | 54,373,518 | 12,266,287 |
| Scribble annotation | 26,909,019 | 3,590,263 | 4,041,995 |
| Scribble rate | 3.40% | 6.60% | 32.95% |

annotations. Owing to difference of size and resolution, the scribble rate of the Zurich Summer dataset is so much higher than that of the other two datasets. The information of these datasets is as follows.

*1) Potsdam Dataset:* Potsdam dataset consists of 38 images with a size of 6000 × 6000 pixels. Its spatial resolution is 0.05 m. In experiments, three bands of red, green, and blue (RGB) are selected from the original four bands IRRGB (infrared, RGB). The class of clutter is ignored for training and testing, and thus, it includes five land cover types: impervious surfaces (imp. surf.), building, low vegetation (low veg.), tree, and car. Among 38 images, 23 images except for the image named 7_10 (removed due to its error annotation) are used for training, and 14 images are used for testing.

*2) Vaihingen Dataset:* Vaihingen dataset comprises 33 aerial images with a resolution of 0.09 m, averaging approximately 2500 × 1900 pixels. The images include IRRG (infrared, red, and green) bands. Similar to the Potsdam dataset, there are five land cover types in Vaihingen dataset: imp. surf., building, low veg., tree, and car. Clutter is also taken as background. To ensure experimental fairness, as specified in the previous works [15], [17], [43], 11 images (IDs: 1, 3, 5, 7, 13, 17, 21, 23, 26, 32, 37) are used for training, and five images (IDs: 11, 15, 28, 30, 34) are used for testing.

*3) Zurich Summer Dataset:* Zurich Summer dataset consists of 20 satellite images captured by QuickBird with a resolution of 0.62 m. The average size of images is 1000 × 1000 pixels, which contains four channels: IRRGB (near-infrared, RGB). Only IRRG bands (infrared, red, and green) are utilized for the experiments. Zurich Summer dataset originally includes eight land cover types: road, building, tree, grass, bare soil, water, railway, and swimming pool (swim. pool). In scribble annotation, railway and road are easily confused with each other; therefore, railway is regarded as road in labeling. According to previous works [15], [17], [43], to ensure experimental fairness, 15 images are used for training, and 5 images (IDs: 16, 17, 18, 19, 20) are used for testing.

TABLE II
RELEVANT INFORMATION OF COMPARATIVE METHODS. "F" DENOTES FULL SUPERVISION AND "S" DENOTES SCRIBBLE SUPERVISION

| Method | Venue | Supervision | Parameters (M) | Flops (B) |
|---|---|---|---|---|
| FCN | CVPR'15 | F/S | 134.3 | 296.3 |
| SegNet | TPAMI'17 | F/S | 29.48 | 170.9 |
| CRGNet | TIP'22 | S | 29.06 | 160.7 |
| MANet | TGRS'22 | F | 35.86 | 77.73 |
| UNetFormer | ISPRS'22 | F | 11.68 | 11.74 |
| ScribFormer | TMI'24 | S | 47.93 | 179.9 |
| AGMM | TPAMI'25 | S | 40.53 | 190.2 |
| SparseFormer | TGRS'25 | S | 42.60 | 168.8 |
| SuperScribble | -- | S | 22.87 | 242.7 |

### B. Comparative Methods and Metrics

In the experiment, in order to highlight the strength of SuperScribble, there are eight representative comparative methods, including FCN [7], SegNet [25], CRGNet [15], MANet [26], UNetFormer [8], ScribFormer [18], AGMM [37], and SparseFormer [17]. FCN and SegNet are classic methods for semantic segmentation. MANet and UNetFormer are used for RSI. CRGNet, ScribFormer, and SparseFormer all introduce dual branches like SuperScribble. In order to clearly identify the gap between SuperScribble and existing FSSS methods, some comparison methods are also trained at full supervision. Table II provides a brief relevant information of comparative methods, where "F" denotes full supervision and "S" denotes scribble supervision. In the training phase, same scribble annotations or full pixelwise annotations are as supervision to train proposed method and comparative methods. In the testing phase, the original full pixel-wise annotations of testing images from the three datasets are used for evaluation. In SuperScribble, the number of parameters for PFE, SFE, and MAF is 10.18, 5.37, and 7.32 M.

To comprehensively compare SuperScribble with the other methods, the experimental results are evaluated based on both visual interpretation and quantitative metrics. Visual interpretation performs by comparing the segmentation results, segmentation details, and error maps of different methods. Quantitative assessment includes $F1$ score of each class, mean $F1$ score (mF1), mean intersection over union (mIoU), and overall accuracy (OA). These metrics collectively evaluate the accuracy between the segmentation result and the ground truth.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ACCURACIES OF DIFFERENT METHODS ON THE POTSDAM DATASET (%). THE PER-CLASS $F1$ SCORE, MF1 SCORE, MIOU, AND OA. BOLD FONTS INDICATE THE BEST RESULTS, FOLLOWED BY UNDERLINED FONTS

| Method | Type | Imp. Surf. | Building | Low Veg. | Tree | Car | mF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|---|
| FCN | | 91.37 | 95.17 | 84.53 | 84.88 | 90.62 | 89.31 | 80.94 | 89.62 |
| SegNet | | 91.35 | 95.13 | 85.25 | 85.23 | 90.71 | 89.59 | 81.36 | 89.88 |
| MANet | FSSS | 93.58 | 96.06 | 87.65 | 88.94 | 96.16 | 92.48 | 86.21 | 91.17 |
| UNetFormer | | **94.12** | **96.75** | **87.78** | **89.03** | **96.31** | **92.80** | **86.79** | **91.59** |
| SuperScribble | | 91.25 | 95.29 | 84.49 | 84.77 | 90.22 | 89.20 | 80.76 | 89.56 |
| FCN | | 81.82 | 85.71 | 66.24 | 67.60 | 71.17 | 74.51 | 60.01 | 76.82 |
| SegNet | | 81.26 | 83.76 | 71.85 | 70.93 | 79.52 | 77.46 | 63.50 | 78.15 |
| CRGNet | | 83.79 | 88.73 | 76.06 | **73.68** | 73.43 | 79.14 | 65.91 | 81.57 |
| ScribFormer | WSSS | 81.56 | 86.82 | 69.60 | 63.97 | 65.93 | 73.58 | 59.03 | 77.11 |
| AGMM | | 81.77 | 80.72 | 66.47 | 65.21 | 71.79 | 73.19 | 58.20 | 75.46 |
| SparseFormer | | 85.31 | 91.47 | 74.56 | 73.34 | 74.01 | 79.74 | 66.95 | 82.42 |
| SuperScribble | | **85.94** | **93.33** | **77.00** | 69.05 | **82.55** | **81.57** | **69.69** | **83.07** |

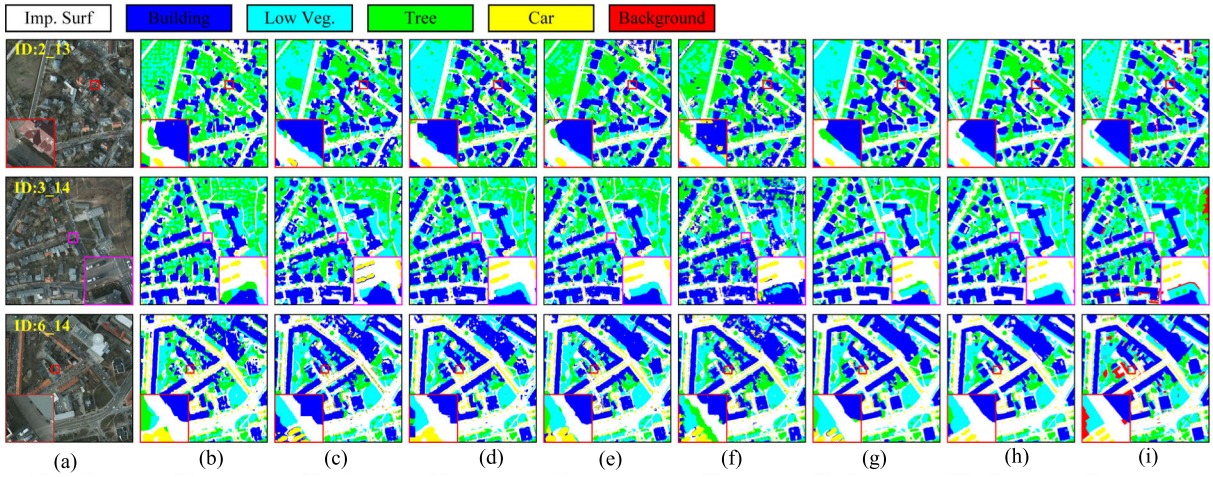



Fig. 9. Results and corresponding details of different WSSS methods on the Potsdam dataset. (a) Image. (b) FCN. (c) SegNet. (d) CRGNet. (e) ScribFormer. (f) AGMM. (g) SparseFormer. (h) SuperScribble. (i) Ground truth.

### C. Comparative Experiments and Analysis

*1) Results on Potsdam Dataset:* Table III shows accuracies of different methods on the Potsdam dataset. The accuracies of FSSS methods are better than WSSS methods, which indicate the importance of training data. In addition, compared with other WSSS methods, the proposed SuperScribble achieves the best mF1, mIoU, and OA at scribble supervision. At full supervision, SuperScribble still exhibits a performance gap compared with other methods, which need to be improved. At scribble supervision, SuperScribble has a certain gap in performance compared to some FSSS methods, especially when it comes to small objects such as trees and cars. But overall, trained with only 3.40% of the fully supervised annotations, SuperScribble's performance is still trustworthy. Due to that the annotated scribble labels don't conform to the prior assumptions of AGMM, the accuracy of AGMM is significantly inferior to that of its performance in natural image semantic segmentation.

Fig. 9 presents the segmentation results and corresponding details of different WSSS methods on the Potsdam dataset. Some methods, such as FCN, SegNet, and AGMM, exhibit fragmented predictions and inaccurate class boundaries, particularly in complex urban areas where buildings, roads, and low vegetation are densely distributed. CRGNet and ScribFormer reduce some misclassification errors but still suffer from noticeable noise and boundary ambiguity, especially around small objects. SparseFormer generates relatively smoother segmentation maps, but it tends to over-smooth object boundaries, resulting in blurred contours and confusion between adjacent categories. Compared with other comparative methods, whether viewed from the overall results or the local details, the proposed SuperScribble achieves more satisfactory visual semantic segmentation results. In Fig. 9, SuperScribble preserves the boundaries of the buildings well. But in upper-right region of image (ID:3_14), SuperScribble also tends to misclassify trees as low vegetation.

*2) Results on Vaihingen Dataset:* Table IV shows accuracies of different methods on the Vaihingen dataset. The proposed SuperScribble achieves an mF1 score of 79.73%, an mIoU of 66.99%, and an OA of 82.22%, which are the best accuracies among all the WSSS methods. However, SuperScribble shows insufficient recognition performance for low vegetation, indicating that further improvement is needed. In addition, compared with Potsdam dataset, as the proportion of training data increases, the gap between SuperScribble and FSSS method UNetFormer has further narrowed from 17.1% to 8.91% on mIoU.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

ACCURACIES OF DIFFERENT METHODS ON THE VAIHINGEN DATASET (%). THE PER-CLASS $F1$ SCORE, MF1 SCORE, MIOU, AND OA. BOLD FONTS INDICATE THE BEST RESULTS, FOLLOWED BY UNDERLINED FONTS

| Method | Type | Imp. Surf. | Building | Low Veg. | Tree | Car | mF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|---|
| FCN | FSSS | 86.97 | 91.81 | 76.16 | 85.45 | 77.29 | 83.54 | 72.18 | 85.18 |
| SegNet | | 88.16 | 92.84 | 74.88 | 84.66 | 82.79 | 84.66 | 73.87 | 85.42 |
| MANet | | 87.58 | 92.47 | 75.44 | 85.18 | 84.27 | 84.99 | 74.29 | 85.48 |
| UNetFormer | | **88.79** | **93.45** | **78.02** | **85.63** | **84.45** | **86.07** | **75.90** | **85.69** |
| SuperScribble | | 87.79 | 92.04 | 76.07 | 85.42 | 80.64 | 84.39 | 73.39 | 85.53 |
| FCN | WSSS | 82.35 | 84.16 | 59.29 | 81.04 | 64.34 | 74.24 | 60.07 | 78.14 |
| SegNet | | 80.34 | 83.72 | 64.38 | 80.82 | 71.04 | 76.06 | 61.90 | 78.03 |
| CRGNet | | 79.65 | 86.58 | 64.55 | 80.59 | 59.60 | 74.20 | 60.03 | 78.21 |
| ScribFormer | | 75.88 | 75.61 | 55.10 | 76.70 | 33.14 | 63.29 | 48.40 | 71.10 |
| AGMM | | 74.10 | 79.05 | 59.07 | 76.94 | 37.51 | 65.33 | 50.35 | 72.48 |
| SparseFormer | | 84.30 | 88.25 | 60.32 | 81.16 | 66.74 | 76.15 | 62.68 | 80.01 |
| SuperScribble | | **84.89** | **89.98** | **69.32** | **82.48** | **71.97** | **79.73** | **66.99** | **82.22** |

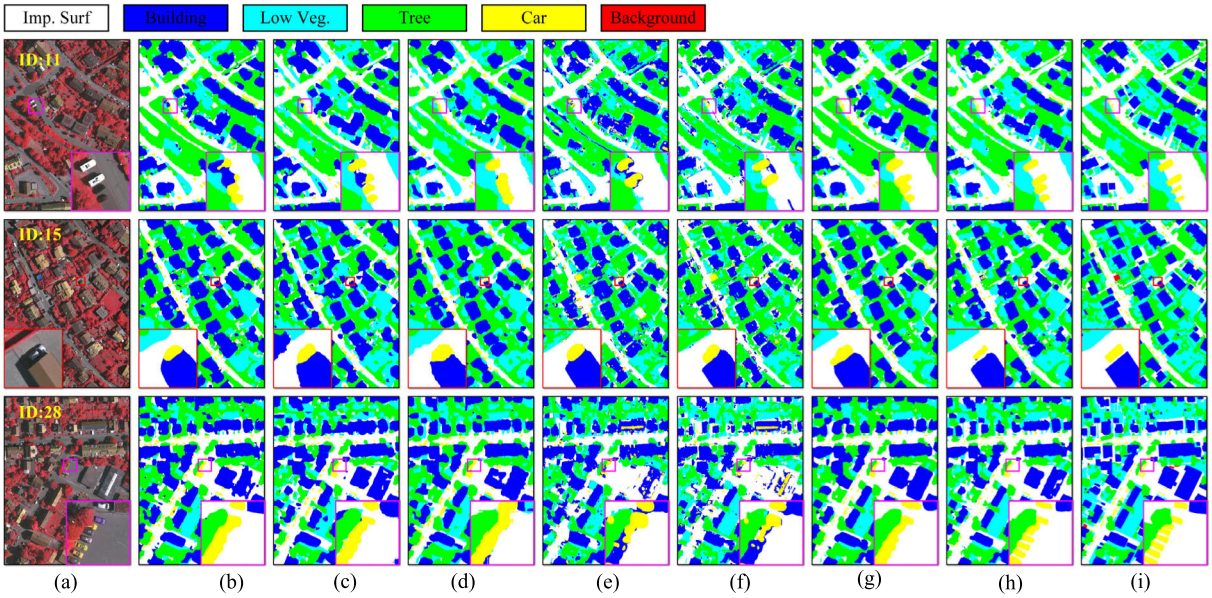



Fig. 10. Results and corresponding details of different WSSS methods on the Vaihingen dataset. (a) Image. (b) FCN. (c) SegNet. (d) CRGNet. (e) ScribFormer. (f) AGMM. (g) SparseFormer. (h) SuperScribble. (i) Ground truth.

Fig. 10 displays the results and corresponding details of different WSSS methods on the Vaihingen dataset. Some methods, such as ScribFormer and AGMM, suffer from fragmented predictions and inaccurate object boundaries, especially in regions with complex spatial structures. CRGNet and SparseFormer could yield better visual results. But they still exhibit noticeable noise and boundary inconsistencies particularly around small objects and object edges. The proposed SuperScribble effectively extracts the main roads. In addition, in terms for details in Fig. 10, SuperScribble could accurately extract small objects, such as road between car and building of the first scene and road between cars of the second scene. Of course, SuperScribble also has certain limitations in Fig. 10, as it is prone to misclassification in shaded regions.

*3) Results on Zurich Summer Dataset:* Table V presents accuracies of different methods on the Zurich Summer dataset. The quantitative evaluation results on it are amazing. With 32.95% of full annotation data, the performance of the WSSS methods can be comparable to that of the FSSS methods. The proposed SuperScribble gains an mF1 score of 92.12%, which is slightly inferior to the UNetFormer in the FSSS methods. Compared with other WSSS methods, SuperScribble shows remarkable performance. It achieves the highest $F1$ scores in six of the seven categories. Moreover, it outperforms other WSSS methods in mIoU by 3.00%–11.02%. However, under scribble supervision, SuperScribble exhibits limited capability in recognizing bare soil, indicating room for improvement.

Because spatial resolution of the Zurich Summer dataset is relatively low, error maps are used for visual interpretation. Fig. 11 shows the results and corresponding error maps of different WSSS methods on the Zurich Summer dataset. In error maps, yellow pixels denote wrong predicted results and blue pixels denote right predicted results. By error maps, it can be observed that results produced by SuperScribble are highly consistent with the ground truth. This demonstrates that the combination of super-pixel features and pixel features is useful in WSSS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

TABLE V

ACCURACIES OF DIFFERENT METHODS ON THE ZURICH SUMMER DATASET (%). THE PER-CLASS $F1$ SCORE, MF1 SCORE, MIOU, AND OA. BOLD FONTS INDICATE THE BEST RESULTS, FOLLOWED BY UNDERLINED FONTS

| Method | Type | Road | Building | Tree | Grass | Bare Soil | Water | Swim. Pool | mF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FCN | | 90.39 | 93.45 | 93.43 | 90.76 | 71.91 | 97.12 | 94.42 | 90.21 | 82.99 | 91.95 |
| SegNet | | 91.83 | **94.12** | 93.73 | 90.42 | 70.40 | 97.85 | 83.90 | 88.89 | 80.98 | 92.41 |
| MANet | FSSS | 91.80 | 93.37 | 92.87 | 91.59 | 73.33 | 98.13 | **95.02** | 90.87 | 84.04 | 92.12 |
| UNetFormer | | **92.21** | 93.96 | 94.45 | **93.39** | 80.68 | 98.39 | 93.97 | **92.44** | **86.33** | **93.63** |
| SuperScribble | | 91.71 | 93.49 | **94.45** | 93.29 | **81.09** | **98.74** | 92.73 | 92.21 | 85.93 | 93.38 |
| FCN | | 86.80 | 90.96 | 90.30 | 85.00 | 65.89 | 93.25 | 93.34 | 86.51 | 77.19 | 88.25 |
| SegNet | | 87.14 | 91.00 | 89.56 | 83.96 | 71.53 | 91.71 | 82.87 | 85.40 | 75.03 | 87.97 |
| CRGNet | | 86.68 | 91.53 | 89.97 | 87.78 | 58.97 | 90.34 | 88.22 | 84.78 | 74.83 | 88.42 |
| ScribFormer | WSSS | 82.34 | 84.47 | 86.43 | 84.40 | 72.87 | 91.16 | **95.24** | 85.27 | 74.89 | 84.64 |
| AGMM | | 89.41 | 93.48 | 94.25 | 93.11 | 78.90 | 90.05 | 85.28 | 89.21 | 80.89 | 91.91 |
| SparseFormer | | 90.77 | 92.99 | 92.66 | 91.01 | 76.58 | 97.17 | 91.05 | 90.32 | 82.85 | 91.95 |
| SuperScribble | | **92.15** | **94.22** | **94.51** | **93.88** | **79.11** | **98.35** | 92.59 | **92.12** | **85.85** | **93.70** |

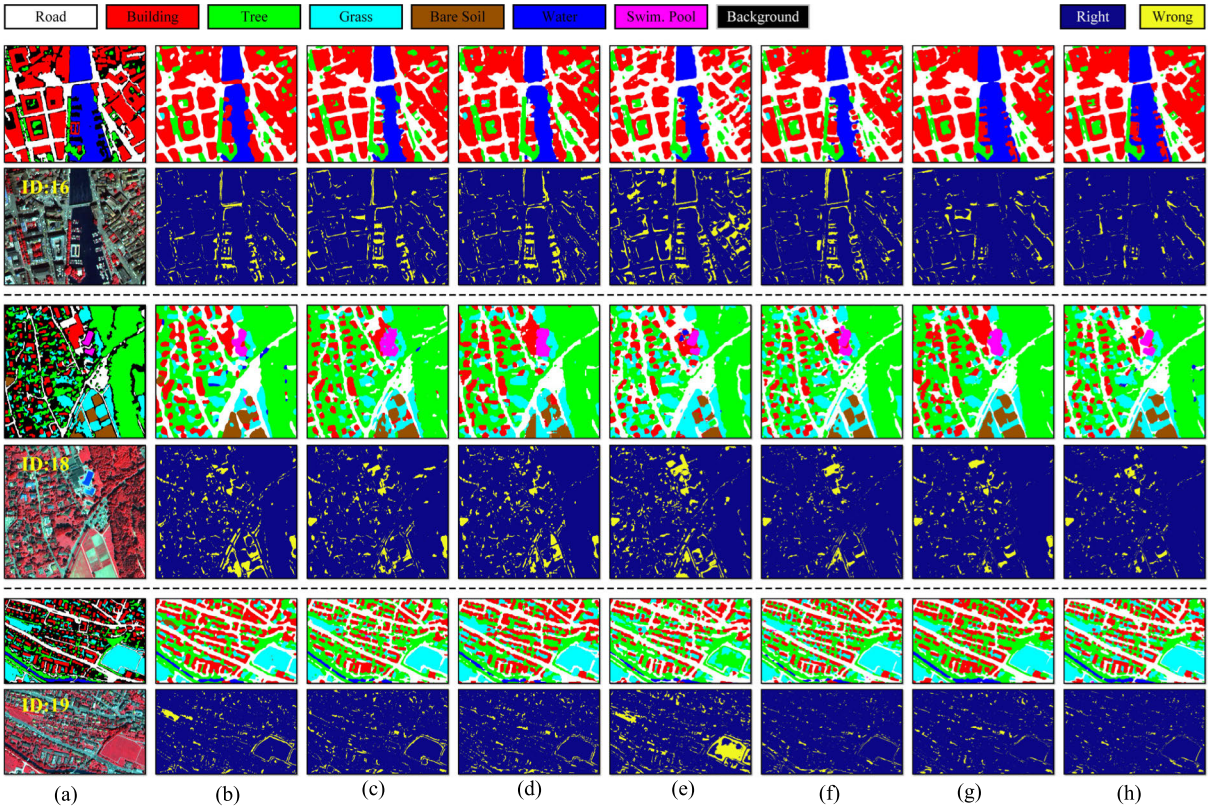



Fig. 11. Results and corresponding error maps of different WSSS methods on the Zurich Summer dataset. (a) Reference. (b) FCN. (c) SegNet. (d) CRGNet. (e) ScribFormer. (f) AGMM. (g) SparseFormer. (h) SuperScribble.

### *D. Ablation Experiments*

The ablation experiments are conducted on the Vaihingen dataset. The effectiveness of PFE, SFE, iterative training, MAF and DCS in SuperScribble are proved. Table VI presents the results of the ablation experiments. In Table VI, "PL" indicates whether models are trained by pseudo labels, "F" in PFE or SFE represents pretrained frozen model. In Table VI, without DCS, the pseudo labels are generated by testing with the pretrained frozen model. Underlined "✓" denotes the model which finally achieves segmentation. In terms of segmentation performance, PFE is superior to SFE, which indicates that the local features are more suitable for the segmentation task. In addition, the data in the third and fourth rows indicate that the pseudo labels enhance the network's ability to extract information. Compared with the data in the third and fourth rows, the data in the fifth and sixth rows are higher, which prove that combination of PFE and SFE is useful. The data in the fifth and sixth rows also emphasize the importance of iterative training. The data in the seventh and eighth rows show that staged training strategy is better for MAF. The data in tenth row highlight the effectiveness of DCS. Fig. 12 shows the segmentation results from PFE, SFE, and SuperScribble. For large objects, such as buildings, SuperScribble could precisely extract contour details. For small objects, such as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

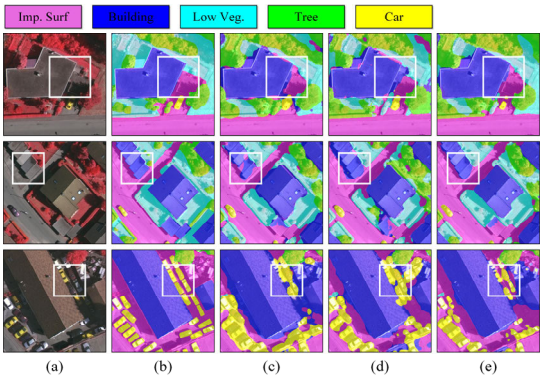



Fig. 12. Results from different models in ablation experiments on the Vaihingen dataset. (a) Image. (b) Ground truth. (c) PFE. (d) SFE. (e) SuperScribble.

TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON THE VAIHINGEN DATASET (%). "PL" INDICATES WHETHER MODELS ARE TRAINED BY PSEUDO LABELS, "F" DENOTES PRETRAINED FROZEN MODEL, AND UNDERLINED "✓" DENOTES THE MODEL WHICH FINALLY ACHIEVES SEGMENTATION

| No. | PFE | SFE | PL | MAF | DCS | mF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|
| 1 | ✓ | | | | | 76.30 | 62.42 | 79.28 |
| 2 | | ✓ | | | | 74.48 | 60.41 | 78.70 |
| 3 | ✓ | | ✓ | | | 78.84 | 65.68 | 81.10 |
| 4 | | ✓ | ✓ | | | 75.47 | 61.63 | 78.82 |
| 5 | ✓ | F | ✓ | | ✓ | 79.54 | 66.72 | 82.00 |
| 6 | F | ✓ | ✓ | | ✓ | 78.33 | 65.11 | 81.23 |
| 7 | ✓ | ✓ | | ✓ | | 75.18 | 61.36 | 79.04 |
| 8 | F | F | | ✓ | | 76.74 | 63.14 | 80.28 |
| 9 | ✓ | ✓ | ✓ | ✓ | | 76.51 | 63.04 | 80.53 |
| 10 | F | F | ✓ | ✓ | | 78.96 | 65.95 | 81.62 |
| Full | F | F | ✓ | ✓ | ✓ | **79.73** | **66.99** | **82.22** |

TABLE VII

INFLUENCE OF DIFFERENT $U_t$ FOR SEMANTIC SEGMENTATION AND PSEUDO LABELS ON THE VAIHINGEN DATASET. "SS" DENOTES SEMANTIC SEGMENTATION ACCURACY, "PL" DENOTES PSEUDO LABELS INFORMATION

| Type | $U_t$ | 0.05 | 0.10 | 0.15 | 0.20 |
|---|---|---|---|---|---|
| SS | mF1 (%) | 79.19 | 79.73 | 79.47 | 79.47 |
| | mIoU (%) | 66.24 | 66.99 | 66.65 | 66.63 |
| | OA (%) | 81.79 | 82.22 | 82.10 | 81.96 |
| PL | mF1 (%) | 84.55 | 83.14 | 82.32 | 81.76 |
| | mIoU (%) | 74.24 | 72.19 | 71.01 | 70.22 |
| | OA (%) | 89.29 | 87.98 | 87.20 | 86.97 |
| | Quantity (M) | 46.70 | 48.43 | 48.44 | 50.15 |

cars, SuperScribble could identify their uniqueness, which avoids incorrectly aggregating multiple small objects into one large object. Fig. 12 indicates that SuperScribble combines the advantages of PFE and SFE, which fuses the local and global information for better segmentation.

In DCS, hyperparameter $U_t$ plays a key role in balancing the trade-off between the quantity and quality of pseudo labels, which in turn affects the overall semantic segmentation performance. Table VII shows how different values of $U_t$ influence semantic segmentation accuracy and pseudo labels accuracy on the Vaihingen dataset. As smaller $U_t$ is, the lower pseudo labels accuracy is, and the more pseudo labels quantity is. When $U_t = 0.1$, the semantic segmentation performance of SuperScribble on the Vaihingen is best.

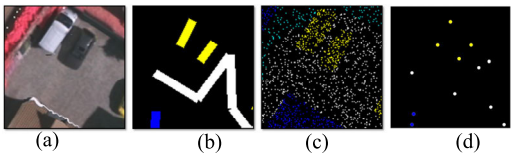



Fig. 13. Annotation data. (a) Image. (b) Scribble. (c) Point annotation, the number of labels is as same as scribble annotation. (d) Corner annotation, constructed from scribble.

## V. DISCUSSION

This article proposes a novel SuperScribble for RSI WSSS with scribble annotation. During the same annotation time, compared with point labels, scribble labels can provide more information. The gap in WSSS performance between point labels and scribble labels is a task worth exploring, which will highlight the potential of scribble labels in practice. In addition, there are also some interesting works to display. For example, the improvement in the quality of the pseudo-labels is worth discussing through DCS, whose generalizability in other models with dual branches is also an important work; therefore, this section focuses on discussions of three aspects.

### A. Point Label Versus Scribble Label

In WSSS, it's meaningful to examine the effectiveness of point annotation versus scribble annotation. In practical scenarios, annotators mainly consider two factors during the labeling process: annotation difficulty and annotation time. Point annotation and scribble annotation exhibit nearly the same level of difficulty, as both only require labeling within the interior of target objects without the need to precisely delineate object boundaries. But as Fig. 13(c) shows, to obtain the same number of point labels as the scribble labels, it will take a much longer time for point annotation. In scribble labeling, scribbles are also composed of points. So, in order to guarantee fairness in labeling time, corner labels from scribble are selected to conduct experiments.

It's found that compared to the point annotation, scribble annotation provides a greater number of valid labels. To further explore its impact on the accuracy of the segmentation, CRGNet, SegNet, and SuperScribble are chosen for evaluation. The segmentation accuracy from different methods trained on different labels on the Vaihingen dataset is displayed in Fig. 14. For all the three methods, trained with scribble annotation yields better segmentation performance than trained with point annotation. This further proves that scribble could provide more information for WSSS, which is promising potential in practical production.

### B. Different Pseudo Labels

As the core of WSSS, generation of pseudo labels is always a hot topic. In this article, the adopted generation strategy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

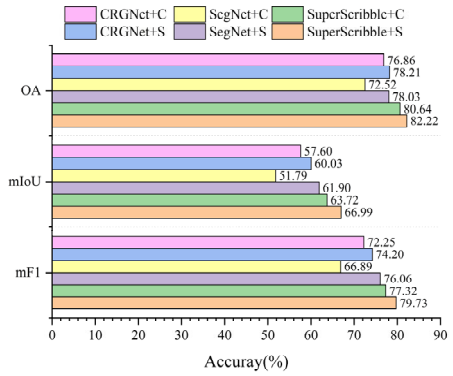



Fig. 14. Segmentation accuracy from different methods trained on different labels on the Vaihingen dataset. "C" denotes corner labels and "S" denotes scribble labels.

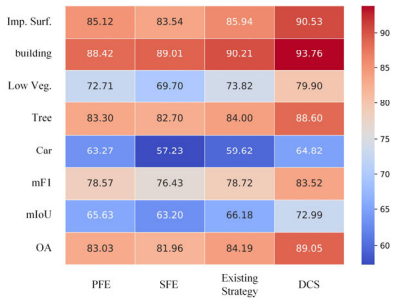



Fig. 15. Accuracies (per-class $F1$, mF1, mIoU, and OA) of different pseudo labels.

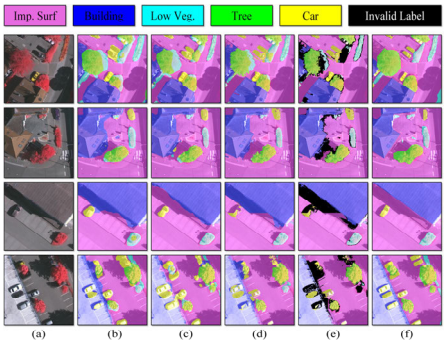



Fig. 16. Pseudo labels generated by different strategies. (a) Image. (b) PFE. (c) SFE. (d) Existing strategy. (e) DCS. (f) Ground truth.

TABLE VIII
EXPERIMENT RESULTS ABOUT GENERALIZABILITY OF DCS ON THE VAIHINGEN DATASET (%)

| Method | W/O DCS | mF1 | mIoU | OA |
|---|---|---|---|---|
| ScribFormer | × | 63.29 | 48.40 | 71.10 |
| | ✓ | 67.30 | 52.24 | 73.51 |
| SparseFormer | × | 76.15 | 62.68 | 80.01 |
| | ✓ | 76.66 | 63.29 | 80.42 |
| SuperScribble | × | 78.96 | 65.95 | 81.62 |
| | ✓ | 79.73 | 66.99 | 82.22 |

DCS, as one of main contributions, is based on the assumption that two independently trained networks with different styles can strengthen the constraints on pseudo labels. In addition, in existing SparseFormer [17], a strategy designed for methods with a dual-branch architecture has been proposed, which determines pseudo labels based on probabilistic uncertainty. Compared with such strategy, the adopted DCS is stricter for pseudo labels, which would rather sacrifice the quantity of pseudo labels in order to guarantee their quality. It is necessary to evaluate the pseudo samples to determine the effectiveness of DCS. Frozen PFE and SFE trained by origin scribble labels are chosen to perform experiments. mF1, mIoU, and OA are used for evaluation, where invalid pixels in pseudo labels do not participate in the assessment.

There are four kinds of pseudo labels, which are generated by PFE, SFE, the strategy in SparseFormer [17], and DCS, respectively. The number of them is 54.884, 54.884, 54.883, and 48.249 M respectively. Fig. 15 presents the accuracies of pseudo labels calculated based on full pixelwise annotations. The idea of DCS is to minimize the interference of incorrect labels on the subsequent network training as much as possible. As Fig. 15 shown, although the number of pseudo labels generated by DCS decreases by 12%, its mIoU is 6.81%–9.79% higher than that of other methods. Fig. 16 displays pseudo labels generated by different strategies. In pseudo label maps, black pixels denote invalid labels. In some areas where segmentation errors are likely to occur, such as near the boundaries of the objects, DCS does not generate pseudo labels. On such areas, pseudo labels generated by other methods might be wrong, which poses challenges for subsequent training.

### C. Generalizability of DCS

The effectiveness of pseudo label generation strategy greatly affects the subsequent segmentation accuracy. The effectiveness of proposed DCS applied in SuperScribble has been proved in ablation experiments. It's easy to find that the proposed DCS could be introduced to some existing methods with dual branches structure, such as ScribFormer and SparseFormer. Therefore, in this section, experiments on generalizability of DCS are conducted. Table VIII shows the effectiveness of DCS in other methods, which further highlights the superiority of DCS in WSSS. Through DCS, the accuracies of ScribFormer's mF1, mIoU, and OA are increased by 4.01%, 3.84%, and 2.41%, respectively. Similar to ScribFormer, the accuracies of SparseFormer and SuperScribble have also been improved through DCS. It shows that introducing DCS in some hybrid dual-branch networks is promising for WSSS.

## VI. CONCLUSION

In this study, a novel WSSS method SuperScribble is proposed, which achieves satisfactory semantic segmentation results in RSI with scribble annotation. SuperScribble consists of three models: SFE, PFE, and MAF. SFE and PFE are designed based on SSN and EfficientNet, respectively, aiming to extract global and local information. MAF further fuses the global super-pixel features and local pixel features from SFE and PFE, respectively, to achieve more accurate segmentation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

results. Through a careful and comprehensive evaluation for the outputs from the SFE and PFE, namely DCS, SuperScribble generates highly reliable pseudo labels, thereby further preventing imprecise labels from misleading the subsequent training.

Experimental results on Potsdam, Vaihingen, and Zurich Summer datasets demonstrate that SuperScribble could achieve excellent results in quantitative metrics compared with other WSSS methods. In visual, SuperScribble performs better in segmenting the boundaries of geographical objects, such as buildings and cars. In Section V, potential application of scribble annotation in WSSS is proved. In addition, the advantage and generalizability of DCS are also demonstrated, which is highly compatible with dual-branch networks Looking ahead. There are some interesting and practical works to do in SuperScribble. On the one hand, as results on Zurich Summer dataset shown, it's not necessary to gain full supervised annotation to train networks, so experiments will be conducted to explore the relationship between the annotation characteristics and segmentation accuracy. On the other hand, DCS will be introduced in more dual-branch networks to further prove its effectiveness. Finally, adopting staged strategy inevitably increases computational costs, so SuperScribble needs to be light-weighted in future work.

## REFERENCES


[1] Y. Dai et al., "Coastal phytoplankton Blooms expand and intensify in the 21st century," *Nature*, vol. 615, no. 7951, pp. 280–284, Mar. 2023, doi: 10.1038/s41586-023-05760-y.

[2] A. Zhao, R. Feng, and X. Li, "ThiefCloud: A thickness fused thin cloud removal network for optical remote sensing image with self-supervised learnable cloud prior," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 11834–11848, Dec. 2025, doi: 10.1109/TCSVT.2025.3585720.

[3] J. Zou, W. He, and H. Zhang, "LESSFormer: Local-enhanced spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535416, doi: 10.1109/TGRS.2022.3196771.

[4] A. Zhao, C. Wang, and X. Li, "A global multiscale hybrid network for hyperspectral image classification," *Remote Sens. Lett.*, vol. 14, no. 9, pp. 1002–1010, Sep. 2023, doi: 10.1080/2150704x.2023.2258467.

[5] Z. Hou et al., "War city profiles drawn from satellite images," *Nature Cities*, vol. 1, no. 5, pp. 359–369, Apr. 2024, doi: 10.1038/s44284-024-00060-6.

[6] H. Wang, N. Wang, and X. Li, "FarmCD: Agricultural information-guided gated network for farmland change detection from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4401714, doi: 10.1109/TGRS.2026.3652586.

[7] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017, doi: 10.1109/TPAMI.2016.2572683.

[8] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022, doi: 10.1016/j.isprsjprs.2022.06.008.

[9] X. Li, L. Xie, C. Wang, J. Miao, H. Shen, and L. Zhang, "Boundary-enhanced dual-stream network for semantic segmentation of high-resolution remote sensing images," *GISci. Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2356355, doi: 10.1080/15481603.2024.2356355.

[10] X. Bai, X. Li, J. Miao, and H. Shen, "A front-back view fusion strategy and a novel dataset for super tiny object detection in remote sensing imagery," *Knowl.-Based Syst.*, vol. 326, Sep. 2025, Art. no. 114051, doi: 10.1016/j.knosys.2025.114051.

[11] Z. Zhang et al., "FlexiSAM: A flexible SAM-based semantic segmentation model for land cover classification using high-resolution multimodal remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 227, pp. 594–612, Sep. 2025, doi: 10.1016/j.isprsjprs.2025.05.028.

[12] Z. Li, X. Zhang, and P. Xiao, "One model is enough: Toward multiclass weakly supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4503513, doi: 10.1109/TGRS.2023.3290242.

[13] S. U. Saeed et al., "Competing for pixels: A self-play algorithm for weakly-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 2, pp. 825–839, Feb. 2025, doi: 10.1109/TPAMI.2024.3474094.

[14] X. Cao, H. Zou, J. Li, X. Ying, and S. He, "OBBInst: Remote sensing instance segmentation with oriented bounding box supervision," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103717, doi: 10.1016/j.jag.2024.103717.

[15] Y. Xu and P. Ghamisi, "Consistency-regularized region-growing network for semantic segmentation of urban scenes with point-level annotations," *IEEE Trans. Image Process.*, vol. 31, pp. 5038–5051, 2022, doi: 10.1109/TIP.2022.3189825.

[16] Y. Cao, X. Huang, and Q. Weng, "A SAM-adapted weakly-supervised semantic segmentation method constrained by uncertainty and transformation consistency," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 137, Mar. 2025, Art. no. 104440, doi: 10.1016/j.jag.2025.104440.

[17] Y. Chen et al., "SparseFormer: A credible dual-CNN expert-guided transformer for remote sensing image segmentation with sparse point annotation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607316, doi: 10.1109/TGRS.2024.3523537.

[18] Z. Li et al., "ScribFormer: Transformer makes CNN work better for scribble-based medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 43, no. 6, pp. 2254–2265, Jun. 2024, doi: 10.1109/TMI.2024.3363190.

[19] Z. Pan, H. Sun, P. Jiang, G. Li, C. Tu, and H. Ling, "CC4S: Encouraging certainty and consistency in scribble-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 8918–8935, Dec. 2024, doi: 10.1109/TPAMI.2024.3415387.

[20] Y. Wei and S. Ji, "Scribble-based weakly supervised deep learning for road surface extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602312, doi: 10.1109/TGRS.2021.3061213.

[21] H. Chen et al., "Structure-aware weakly supervised network for building extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412712, doi: 10.1109/TGRS.2022.3217830.

[22] R. Yan et al., "Global–local semantic interaction network for salient object detection in optical remote sensing images with scribble supervision," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3360106.

[23] H. Zhang, M. Lin, G. Yang, and L. Zhang, "ESCNet: An end-to-end superpixel-enhanced change detection network for very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 1, pp. 28–42, Jan. 2023, doi: 10.1109/TNNLS.2021.3089332.

[24] Y. Chen et al., "Superpixel-aware credible dual-expert learning for land cover mapping using historical land cover product," *ISPRS J. Photogramm. Remote Sens.*, vol. 223, pp. 296–316, May 2025, doi: 10.1016/j.isprsjprs.2025.02.014.

[25] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017, doi: 10.1109/TPAMI.2016.2644615. t

[26] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713, doi: 10.1109/TGRS.2021.3093977.

[27] M. Sultana, A. Mahmood, T. Bouwmans, M. H. Khan, and S. K. Jung, "Moving objects segmentation using generative adversarial modeling," *Neurocomputing*, vol. 506, pp. 240–251, Sep. 2022, doi: 10.1016/j.neucom.2022.07.081.

[28] P. J. Navarro, E. Salas, F. Rosique, B. Ruszczak, and M. Egea-Cortines, "Conditional generative adversarial network for segmentation and image reconstruction of grape berries in variable illumination conditions," *Expert Syst. Appl.*, vol. 287, Aug. 2025, Art. no. 128076, doi: 10.1016/j.eswa.2025.128076.

[29] A. Li, X. Li, and X. Ma, "Residual dual U-shape networks with improved skip connections for cloud detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2023.3337860.

[30] Y. Cheng et al., "Multi-scale feature fusion and transformer network for urban green space segmentation from high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103514, doi: 10.1016/j.jag.2023.103514.

[31] Y. Liu et al., "A transformer-based multi-modal fusion network for semantic segmentation of high-resolution remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104083, doi: 10.1016/j.jag.2024.104083.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.

[32] B. Zhang, X. Gao, S. Yu, and W. Liu, "Enhanced online CAM: Single-stage weakly supervised semantic segmentation via collaborative guidance," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110787, doi: 10.1016/j.patcog.2024.110787.
[33] D. Lin, J. Dai, J. Jia, K. He, and J. Sun, "ScribbleSup: Scribble-supervised convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3159–3167, doi: 10.1109/CVPR.2016.344.
[34] Y. Li, C. Hu, K. Ren, H. Xi, and J. Fan, "UASeg: Uncertainty aware weakly supervised semantic segmentation for autonomous driving," *Expert Syst. Appl.*, vol. 292, Nov. 2025, Art. no. 128609, doi: 10.1016/j.eswa.2025.128609.
[35] S. Liu, S. Zhou, Y. Wang, K. Lu, W. Liu, and Z. Wang, "Uncertainty-guided weakly supervised segmentation of cardiac substructures with adapter fine-tuning and Fourier feature extraction," *Expert Syst. Appl.*, vol. 283, Jul. 2025, Art. no. 127583, doi: 10.1016/j.eswa.2025.127583.
[36] J. Ahn, S. Cho, and S. Kwak, "Weakly supervised learning of instance segmentation with inter-pixel relations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2204–2213, doi: 10.1109/CVPR.2019.00231.
[37] L. Wu et al., "Modeling the label distributions for weakly-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6290–6306, Aug. 2025, doi: 10.1109/TPAMI.2025.3557047.
[38] Y. Lu, J. Zhang, D. Kong, X. Zheng, and X. Wang, "BACF: Boundary-aware collaborative framework for weakly supervised semantic segmentation," *Pattern Recognit.*, vol. 174, Jun. 2026, Art. no. 112989, doi: 10.1016/j.patcog.2025.112989.
[39] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.
[40] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," 2018, *arXiv:1807.06521*.
[41] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 363–380, doi: 10.1007/978-3-030-01234-2_22.
[42] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012, doi: 10.1109/TPAMI.2012.120.
[43] Y. Hua, D. Marcos, L. Mou, X. X. Zhu, and D. Tuia, "Semantic segmentation of remote sensing images with sparse annotations," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3051053.



**Anqi Zhao** received the B.S. degree in remote sensing science and technology from Wuhan University, Wuhan, China, in 2024, where he is currently pursuing the M.S. degree with the School of Remote Sensing and Information Engineering.

His research interests include computer vision, image processing, and semantic segmentation.



**Hongjin Wang** received the B.E. degree in spatial information and digital technology from Wuhan University, Wuhan, China, in 2024, where he is currently pursuing the M.S. degree with the School of Remote Sensing and Information Engineering.

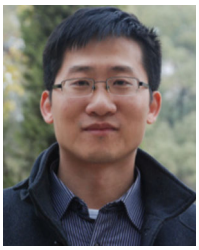
His research interests include deep learning, remote sensing image processing, and change detection.



**Xinghua Li** (Senior Member, IEEE) received the Ph.D. degree in cartography and geographical information engineering from Wuhan University, Wuhan, China, in 2016.

In 2016, he joined the School of Remote Sensing and Information Engineering, Wuhan University, as a Professor. He is the PI of two National Natural Science Foundation of China (NSFC) projects and two subprojects of the National Key Research and Development Program of China. He has published more than 100 peer-reviewed articles. His research interests focus on multitemporal remote sensing image processing and applications, and deep learning.

Dr. Li won the Best reviewer of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2021 and *International Journal of Applied Earth Observation and Geoinformation* in 2021. He has served as a reviewer for more than 90 SCI-indexed journals. He is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *Journal of Image Science and Technology*, an Editorial Member of *Remote Sensing*, and a Guest Editor of *Journal of Applied Remote Sensing*.



**Huanfeng Shen** (Senior Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

In 2007, he joined the School of Resource and Environmental Sciences (SRES), Wuhan University, where he is currently a Luojia Distinguished Professor and the Associate Dean of SRES. He was or is a Principal Investigator of two projects supported by the National Key Research and Development Program of China and six projects supported by the National Natural Science Foundation of China. He has authored over 100 research articles in peer-reviewed international journals. His research interests include remote sensing image processing, multisource data fusion, and intelligent environmental sensing.

Dr. Shen is a Council Member of China Association of Remote Sensing Application, an Education Committee Member of Chinese Society for Geodesy Photogrammetry and Cartography, and a Theory Committee Member of Chinese Society for Geospatial Information Society. He is currently a member of the Editorial Board of *Journal of Applied Remote Sensing, Journal of Geography*, and *Journal of Geo-Information Science*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:45:04 UTC from IEEE Xplore. Restrictions apply.